

Sample Paper 4 — Answer Key

Total: 40 marks (30×1 mark + 5×2 marks), 60 minutes.

Section 1 — Logical Reasoning

1. **D) Ball** — Hammer, Saw, Screwdriver are tools; a Ball is not.
2. **A) Ship** — A pilot commands an airplane; a captain commands a ship.
3. **A) VKIGT** — The rule is "+2 letters": $T \rightarrow V$, $I \rightarrow K$, $G \rightarrow I$, $E \rightarrow G$, $R \rightarrow T$.
4. **C) 35** — The pattern adds 7 each time: 7, 14, 21, 28, 35.
5. **B) 5th** — Position from right = total – position from left + 1 = $7 - 3 + 1 = 5$.
6. **A) North** — Facing West, a right turn faces you North.
7. **B) Thursday** — $15 - 1 = 14$ days later = exactly 2 weeks later = same day, Thursday.
8. **D) 29** — 21, 24, 27 are all multiples of 3; 29 is not.
9. **C) 243** — Each term is multiplied by 3: 3, 9, 27, 81, 243.
10. **A) ICOG** — The rule is "+2 letters" (with Y wrapping to A): $G \rightarrow I$, $A \rightarrow C$, $M \rightarrow O$, $E \rightarrow G$.

Section 2 — Mathematical Reasoning

11. **A) 12134** — $8456 + 3678 = 12134$.
12. **A) 5433** — $10000 - 4567 = 5433$.
13. **A) 2048** — $512 \times 4 = 2048$.
14. **A) 672** — $6048 \div 9 = 672$ exactly.
15. **B) 12** — 12 is divisible by both 3 and 4. (9 isn't divisible by 4; 16 isn't by 3; 20 isn't by 3.)
16. **B) 37** — 37 has no factors besides 1 and itself. ($33=3 \times 11$, $39=3 \times 13$, $42=2 \times 21$.)
17. **C) $\frac{3}{5}$** — $\frac{3}{5} = 0.6$.
18. **A) 7 L 500 mL** — $7500 \text{ mL} = 7000 \text{ mL} + 500 \text{ mL} = 7 \text{ L } 500 \text{ mL}$.
19. **B) 33 cm** — An equilateral triangle has 3 equal sides: $3 \times 11 = 33 \text{ cm}$.
20. **B) 3** — A triangle has 3 vertices (corners).

Section 3 — Everyday Mathematics

21. **A) 2330** — $1350 + 980 = 2330$.
22. **C) 360** — $24 \times 15 = 360$.
23. **B) 72** — $504 \div 7 = 72$.

24. **B) 3 h 35 min** — 2:35 p.m. to 6:35 p.m. would be 4 h, but the bus arrives 25 minutes earlier at 6:10 p.m., so the trip is $4 \text{ h} - 25 \text{ min} = 3 \text{ h } 35 \text{ min}$.
25. **B) 20** — $15 \text{ L} = 15000 \text{ mL}$; $15000 \div 750 = 20$ bottles.
26. **B) ₹128** — $(2 \times ₹45) + ₹38 = ₹90 + ₹38 = ₹128$.
27. **C) 200 L** — $1/6 \times 600 = 100 \text{ L}$ per hour; in 2 hours = 200 L.
28. **B) 41 kg 250 g** — $22500 \text{ g} + 18750 \text{ g} = 41250 \text{ g} = 41 \text{ kg } 250 \text{ g}$.
29. **C) 18** — $252 \div 14 = 18$ rows.
30. **B) ₹1,050** — $₹1200 - ₹150 = ₹1050$.

Section 4 — Achievers Section (2 marks each)

31. **C) 182** — Number = $(11 \times 16) + 6 = 176 + 6 = 182$.
32. **B) 720** — Let larger = L, smaller = S. $L + S = 1620$, $L - S = 180$. Adding: $2L = 1800 \rightarrow L = 900$, so $S = 720$.
33. **C) ₹2,800** — To brother = $1/5 \times 4500 = ₹900$; remaining = ₹3600; to sister = $2/9 \times 3600 = ₹800$; left = $3600 - 800 = ₹2800$.
34. **C) 53** — Yellow = 5; green = $5 + 7 = 12$; red = $3 \times 12 = 36$; total = $5 + 12 + 36 = 53$.
35. **C) 36** — These are square numbers ($1^2, 2^2, 3^2, 4^2, 5^2, 6^2, 7^2$); the missing one is $6^2 = 36$.
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Scoring guide (out of 40): 32+ is Level-2 cutoff range — strong. 24–31 is solid Level-1 performance — review the missed sections. Below 24 — revisit the relevant topics before attempting Sample Paper 5.
